

Library overview

Note that the stack comments added is the stack while loading the mentioned source code chapter. Stack comments explanation, there could be a number behind any of the stack comment tokens! +n +n1 +n2 etc.

name = String to be parsed from the input stream
ccc<ch> = String to be parsed delimited by the character 'ch'
a = Address
a b c etc. = Anonymus stack items
ch = ASCII character number
d = Double number
+n = Positive number
n = Signed number
u = Unsigned number
f = Flag (0 or -1)
pin = GPIO pin number
i*x j*x = Undefined stack contents

Tools

Tools function	Description
COPYRIGHT (--)	Copyright message
VERSION (--)	Library version info
UART (--)	Switch back to UART interface
TRY ADD (--)	Two pseudonyms for NEED & RUN
RESTORE-LIB (--)	Restore end of tools library pointer
OPEN-LIB (--)	Words to extend this library
WIPE-LIB (--)	Words to extend this library
CLOSE-LIB (--)	Words to extend this library
CHAPTERS (--)	Words to extend this library
ALPHA (--)	Show library in alphabetic order
-CHAPTERS (--)	Show library in reverse order
NEED((--)	Multi name version of NEED
EXAMINE (--)	Show source code belonging to "name"

LOOK-AT (--)	Show source code belonging to "name"
.LIBS (--)	Show all available libraries
.LIB (--)	Show active library
PIN (pin --)	Change 'pin' number for S?
48MHZ (--)	Change clock to 48 MHz
125MHZ (--)	Change clock to 125 MHz
132MHZ (--)	Change clock to 132 MHz
250MHZ (--)	Change clock to 250 MHz
38K4 (--)	Change baudrate to 38400 baud
115K2 (--)	Change baudrate to 115200 baud
460K8 (--)	Change baudrate to 460800 baud
921K6 (--)	Change baudrate to 921600 baud
BAUD (u --)	Change baudrate to baudrate u from stack
-ROT (--)	Rotate top item to third on the stack
ROLL (--)	Move x-th item on the stack to the top
2TUCK (--)	Copy d2 below d1
2ROT (--)	Rotate third item to top of the stack
-2ROT (--)	Rotate top item to third on the stack
ON OFF (--)	Set or clear the cell at 'addr'
2! (--)	Store the double dx at 'addr'
2@ (--)	Read the double dx from 'addr'
0> (--)	'f' is true when n is signed greater then zero
ARSHIFT (--)	Arithmetic right shift by +n positions
D- (--)	Subtract d2 from d1 leaving d3
M+ (--)	Add n to d1 leaving d2
DLSHIFT DRSHIFT (--)	Double logical left & right shift
SM/REM (--)	Symetric division with remainder
2LOG (--)	Calculate binary logarithm y of u
CHARS (--)	Add character address calculation
ERASE (--)	Erase +n bytes from addr
J K (--)	Nested loop indexes
PAD (--)	Scratch pad area of 32 bytes
SOURCE (--)	Input source manipulation words
CASE (--)	The CASE statement word set
UNUSED (--)	Leave free dictionary space

WORD (--)	Parse the string delimited by ch leave it at a
[COMPILE] (--)	Compile the word behind it always
ABORT" (--)	Error message with inline string
[IF] (--)	Interactive control structure word set
FORTH-WORDLIST (--)	Compilation wordlist manipulation
GET-ORDER (--)	Manipulate search order word set
SEARCH-WORDLIST (--)	Search for wordname in given wordlist id
-TRAILING (--)	Cut trailing spaces from a string
.S (--)	Non-destructive display of data stack
[DEFINED] (--)	Check if a word exist
[UNDEFINED] (--)	Check if a word not exist
CRC (--)	RP2040 style CRC generator
RANDOM (--)	32-bit pseudo random Marsaglia generator, including CHOOSE (u1 – u2)
THENS (--)	Close all open IFs
.BYTE (--)	Print byte number in hexadecimal
.HEX (--)	Print number unsigned in hexadecimal
PCHAR (--)	Convert all data to printable characters
MANY (--)	Redo current input line until a key was hit
STOP? (--)	Leave flag when a key was pressed, hold on a space, abort on Esc.
RECUR (--)	For use with for-next, it manipulates the index using keys from the keyboard
DMP (--)	Simple dump routine, needs address only
DUMP (--)	Classic Forth dump tool
WORDS (--)	Show words in top vocabulary
@NAME (--)	Read counted string from a header
>NFA (--)	Try to convert an address to name field address
?TEXT (--)	Search for inline text string
?HEAD (--)	Check for a valid header at given address
.DATA (--)	Print data word as chars and hexadecimal
SEE (--)	noForth decompiler
.VOCS (--)	Show all present vocabularies
.SHIELDS (--)	Show all present shields
TOOLS\ (--)	Add basic noForth tool set

LARGE-TOOLS (--)	Add extended noForth tool set
ALLWORDS (--)	16 times WORDS in noForth using kangaroo method
MEMMAP (--)	Show noForth memory map
.CFG (--)	Show noForth configuration
LAST\ (--)	Remove code behind the last present shield
-FLY (--)	Make all control structures interactive
COMMACODE (--)	Build assembler less code definitions
DAS\ (--)	Complete RP2040 disassembler
ASM\ (--)	noForth RP2040 T(humb) assembler
+ASM\ (--)	RP2040 assembler extension
PIO\ (--)	RP2040 PIO (dis)assembler v2
PIOBASE\ (--)	Minimal PIO control wordset v2
-LITERAL (--)	Literal compiler, base addr. and offset
\$VARIABLE (--)	String manipulation word set
-TAIL -HEAD (--)	Cut characters from a string
BITARRAY (--)	Bit array word set, for compact on/off noting and operators to handle this array
*COPY (--)	Copy any bit array to another, etc.
COUNT* (--)	Count the bits set in given bit array
*UP? (--)	Get the bit number of the lowest bit set & true and clear that bit. When no bit set is found leave false
(? (--)	Debugging through run-time stack check
?TASK (--)	Check if a task is valid
TASK (--)	Add, start & change background tasks
TASKS (--)	Multitasker tools wordset
LOCK (--)	Semaphores, tools for sharing devices
TASKER\ (--)	Add complete multitasker wordset
SET-FREQ (--)	Set CPU frequency from 12MHz to 400MHz
TESTER\ (--)	Add adapted Hayes test suite
TRACER\ (--)	Very basic number tracer with independent number conversion (.B .DUMP etc.)
TURNKEY (--)	Generate Intel-Hex from current boot image It can be converted to a UF2 with a small Win32Forth program
MEASURE\ (--)	Measure timing of code parts

Hardware

Hardware function	Description
CORE\ (--)	Add inter core communication
CORE1\ (--)	A blinker running on core-1
CORE0 (--)	A counter running on core-0
CORE1 (--)	A counter display program running on core-1
BUTTON (--)	A switch running on one core
BLINK (--)	Simple GPIO25 flashing LED demo
ADC (--)	Using the onboard ADC
BOOTKEY? (--)	Reuse BOOTSEL switch for Forth input
TEMPERATURE (--)	Use the built-in temperature sensor demo
PWM-ON (pin --)	Onboard dual PWM usage example on GPIO & GPIO + 1
ALARM (--)	Using the alarm registers as interval timer
KHZ> (--)	Convert spi-clock to divider values
LOOPBACK (--)	Loopback mode for spi-0 on/off
SPI\ (rx-pin 0 1 --)	spi-0 or spi-1 driver
I2C\ (clock sda-pin --)	Built-in I2C on all clock speeds
I2C-SLAVE\ (sda-pin --)	I2C slave implementation, needs wanted SDA pin number on the stack
IO-SLAVE (--)	I2C PCF8574 style slave driver
COUNTER (--)	I2C slave demo (master code part) set clock frequency & SDA pin number on the stack
MDMP (--)	I2C slave demo (master code part) set clock frequency & SDA pin number on the stack
MEM-SLAVE (--)	Implements an I2C memory slave, shares a memory part over the I2C-bus
>PCF8574 (--)	PCF8574 driver primitives
RUNNER (--)	A few PCF8574 demo drivers
24C02\ (--)	Small I2C-EEPROM code example
EEPROM (--)	An EEPROM code example
DEV? (--)	Check if a device is present on the I2C-bus
SCAN-I2C (--)	An i2c bus scanner
PCF8591\ (--)	I2C analog input & output using a PCF8591
LM75\ (--)	I2C temperature measuring with LM75

Hardware function	Description
WS2812 (--)	Control two WS2812 LEDs separately
WS2812START\ (--)	Self starting dual WS2812 PIO driver
WS2812MULTI\ (--)	Self starting dual WS2812 PIO driver using the multitasker
(--)	Read & compile 16-bit character row
(--)	Read & compile 8-bit character row
'GRAPH (--)	Graphic characters of 8x4 pixels
'SMALL (--)	Small characters set of 8x5 bits
'THIN (--)	A 16x6 thin character set
'BOLD (--)	A 16x8 bold character set
{OLED (--)	OLED driver for GPIO16 and SPI-0
OLED-SPI\ (--)	Primitive SPI OLED text driver
I2C-OLED\ (--)	Primitive I2C OLED text driver
&CR (--)	Add a scrolling display redefining the words &CR & &PAGE
THIN (--)	Output 16x6 thin characters
BOLD (--)	Output 16x8 bold characters
SMALL (--)	Output 8x5 small characters
GRAPHIC (--)	Output 8x4 graphic characters
SMALL-DEMO (--)	Small character scrolling demo
SCROLL-DEMO (--)	Thin character scrolling demo
MIXED-DEMO (--)	Mixed character size scrolling demo
THIN-DEMO (--)	Large thin character size scrolling demo
BOLD-DEMO	Large bold character size scrolling demo
GRAPHIC-DEMO (--)	Display graphic character set
EGEL-DEMO (--)	Moving hedgehog demo
SPI-OLED-DEMO\ (--)	Add all demos for an SPI OLED screen
I2C-OLED-DEMO\ (--)	Add all demos for an I2C OLED screen
OLED-APP\ (--)	Self starting OLED app
ST7789\ (pin --)	240 x 280 TFT driver using SPI from pin
BAMBOE\ (pin #bamboe --)	Add BAMBOE driver on pin for #bamboe's

PIO

PIO function	Description
BIT-TOGGLE1 (sm pio pin --)	Simple but flexible bit toggle
BIT-TOGGLE2 (pin --)	More advanced bit toggle on SM0 & PIO0
BIT-TOGGLE3 (pin --)	Idem 2 but with use of side-set & set
IN&OUT1 (pin --)	Use GPIO to activate GPIO+1
IN&OUT2 (pin --)	Toggle GPIO+1 using MOV, only
IN&OUT3 (pin --)	Toggle GPIO+1 using WAIT, & MOV,
IRQ-1 (pin --)	Two communicating programs using IRQ
IRQ-2 (pin --)	Idem 1, but using wrap function
MUSIC-0 (pin --)	Frequency generation using 4 GPIO's on 4 SM's uses clone
ON&OFF-1 (pin --)	Example of PIN? and MOV, toggles GPIO+1
ON&OFF-2 (pin --)	Example of PIN? and MOV, GPIO+1 on/off
ON&OFF-1 (pin --)	Idem 2, example of PIN? WAIT, and MOV,
PWM-1 (sm pio pin --)	1000 Hz PWM, range 0 to 100
PWM-2 (sm pio pin --)	1000 Hz PWM, range 0 to 200
PWM-3 (sm pio pin --)	1000 Hz PWM, range 0 to 400
PWM-4 (sm pio pin --)	10000 Hz PWM, range 0 to 400
ROTARY-0 (pin --)	Rotary encoder on GPIO to GPIO+3
ENCODER-DEMO (--)	Encoder demo on GPIO26 to GPIO29
SPI-0 (sm pio clock pin --)	SPI I/O using three GPIO pins, clock in Khz 8 data bits and autopull & autopush
SPI-1 (sm pio clock pin --)	SPI I/O using four GPIO pins, clock in Khz 8 data bits, PULL, and autopush
SPI-2 (sm pio clock pin --)	SPI I/O using four GPIO pins, idem 2 but slightly different Forth usage
SPI-3 (sm pio clock pin --)	SPI I/O using four GPIO pins, the chip select is done by the Forth program
SPI-4 (sm pio clock pin --)	SPI out using four GPIO pins
SPI-5 (--)	SPI I/O using GPIO26 to GPIO29
UART-0 (--)	115K2 UART output on GPIO26
UART-1 (--)	Dual UART output on GPIO26 = 115K2 & GPIO27 = 38K4 using clone
UART-2 (--)	115K2 UART output on SM0 and GPIO26 & input on on SM1 and GPIO27

PIO function	Description
UART-3 (--)	460K8 UART TX and RX on GPIO26 & GPIO27 adding a simple chat function
UART-4 (--)	460K8 alternative UART for noForth
UART-5 (--)	9600 baud alternative UART for noForth
WS2812-0 (pin --)	Single WS2812 driver at GPIO from stack
WS2812-1 (sm pio pin --)	Single WS2812 driver demo & GPIO25 led flasher on two state machines
WS2812-2 (--)	Multiple WS2812 driver demo on GPIO23 & GPIO28, on two state machines using clone
WS2812-3 (--)	Multiple WS2812 LED on GPIO28 alternative driver demo, with Forth demo program & special functions
WS2812-4 (--)	Multiple WS2812 LED on GPIO28 precompiled PIO-program, with Forth demo program & special functions
CARS\ (--)	Highway simulation on a string of WS2812 leds (runs in the multitasker!)
DHT22\ (--)	DHT22 humidity & temperature sensor on GPIO5
PIO-US\ (sm pio gpio --)	Ultrasonic sensor driven by PIO, can be used on all sensors with a HC-SR04 like protocol
SHOW-US (--)	Test program for PIO-US\
>BAMBOE (sm pio gpio --)	Generic PIO bamboe driver for 1 to 8 chained bamboe's
PSERVO\ (sm pio --)	Driver for maximal 10 servo's connected to GPIO4 to GPIO13
MOVES (--)	Test program on GPIO12 & GPIO13 for the PSERVO\ driver
PIOSERVO\ (sm pio gpio +n --)	Driver for 1 to 4 servo's (+n) using as many state machines